

# HARDIK KUMAR SINGH

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## EXPERIENCE

### Grid Dynamics

Software Engineer

Bengaluru, India

Jan 2025 – Present

#### Leading Global Card Network

- Engineered Java backend enhancements across merchant management workflows, delivering production features, defect resolutions, and business-driven platform improvements.
- Resolved a merchant data synchronization defect by integrating post-onboarding update flows with downstream processing, eliminating stale reporting data through cross-team collaboration.
- Enhanced API contracts with **three validation scenarios**, replacing generic HTTP 400 responses with descriptive error enums to improve merchant integration and diagnostics.
- Proposed and introduced clearer Virtual Terminal terminology, aligning merchant-facing labels with business functionality to reduce ambiguity.

#### AI-Powered Talent Acquisition Platform

- Developed backend systems for an **AI-driven Talent Acquisition Platform**, reducing manual candidate screening effort by **90%**.
- Built asynchronous REST APIs using **FastAPI** and integrated **Google Vertex AI** with **OpenAI GPT-4** for semantic CV-to-JD matching, improving recruiter efficiency by **35%** and matching accuracy by **35%**.
- Designed an autonomous candidate scoring engine achieving a **91% reduction in screening time** while improving shortlisting consistency by **30%**.
- Containerized backend services using **Docker** and deployed on **Google Cloud Run** with **Cloud SQL (PostgreSQL)**, reducing operational overhead by **25%**.

#### LLM-Powered Learning Assistant

- Led backend development of an **LLM-based Learning Assistant**, transforming **90%** of diverse learning content into interactive learning experiences.
- Built **35+ REST APIs** using **FastAPI**, integrating **PostgreSQL**, **Redis**, and **ChromaDB**, while implementing secure code execution using **Firecracker microVMs**, reducing execution-level security risks by **80%**.
- Integrated AI chatbot, Q&A generation, and automated code evaluation with cross-functional teams, increasing learner engagement by **30%**, while containerizing services using **Docker** for consistent deployments.

## PROJECTS

### OptiVision | Python, TensorFlow, OpenCV, Jetson Nano

Jan 2024 – May 2024

- Developed a real-time multi-camera object detection pipeline using **CenterNet ResNet-101**, achieving **32.4 mAP** with **34 ms** inference latency on Jetson Nano.
- Processed **5+ concurrent RTSP streams** for multi-object detection and tracking, achieving **85%+ detection accuracy** while maintaining real-time performance.
- Established an event logging pipeline capturing **10k+ object detections** to support downstream analytics and debugging.
- Implemented a **TF-IDF**-based search assistant for querying historical detection events without manual log inspection.

## TECHNICAL SKILLS

**Programming Languages:** Java, C++, SQL

**Backend Development:** Spring Boot, FastAPI, REST APIs, Microservices, Distributed Systems

**Databases:** PostgreSQL, MySQL, MongoDB, Redis, ChromaDB

**Messaging & Streaming:** Apache Kafka

**Cloud & DevOps:** Google Cloud Platform (Cloud Run, Cloud SQL), Amazon Web Services (EC2), Docker, Kubernetes

**AI Technologies:** Google Vertex AI, OpenAI GPT-4

**Developer Tools:** Git, GitHub, GitLab, Maven, JUnit, Mockito, IntelliJ IDEA, Visual Studio Code

**Methodologies:** Agile, Scrum

## CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)



## EDUCATION

Motilal Nehru National Institute of Technology, Allahabad

B.Tech in Computer Science and Engineering

CPI: 8.27

Dec 2021 – May 2025